Lab 6: Code and Branch Coverage Analysis

H K Manaswin | PES1UG24AM108 | Program: L6_L7_42 Salon Booking

Repository: https://github.com/SETAPESU26/L6_L7_42_salon_booking

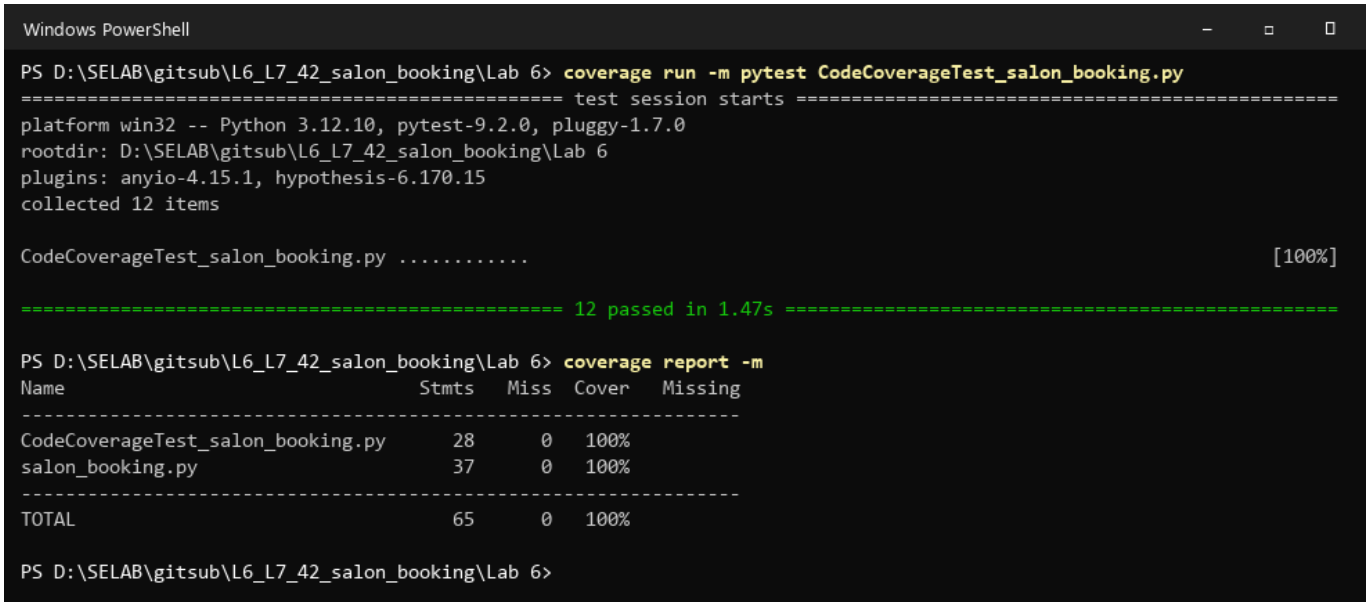
Summary

Test file	Tests	Line coverage of salon_booking.py	Branch coverage of salon_booking.py
test_salon_booking.py (starter)	2	38%	32%
CodeCoverageTest_salon_booking.py	12	100%	98% (missed 21->23)
BranchCoverageTest_salon_booking.py	24	100%	100% (26/26 branches)

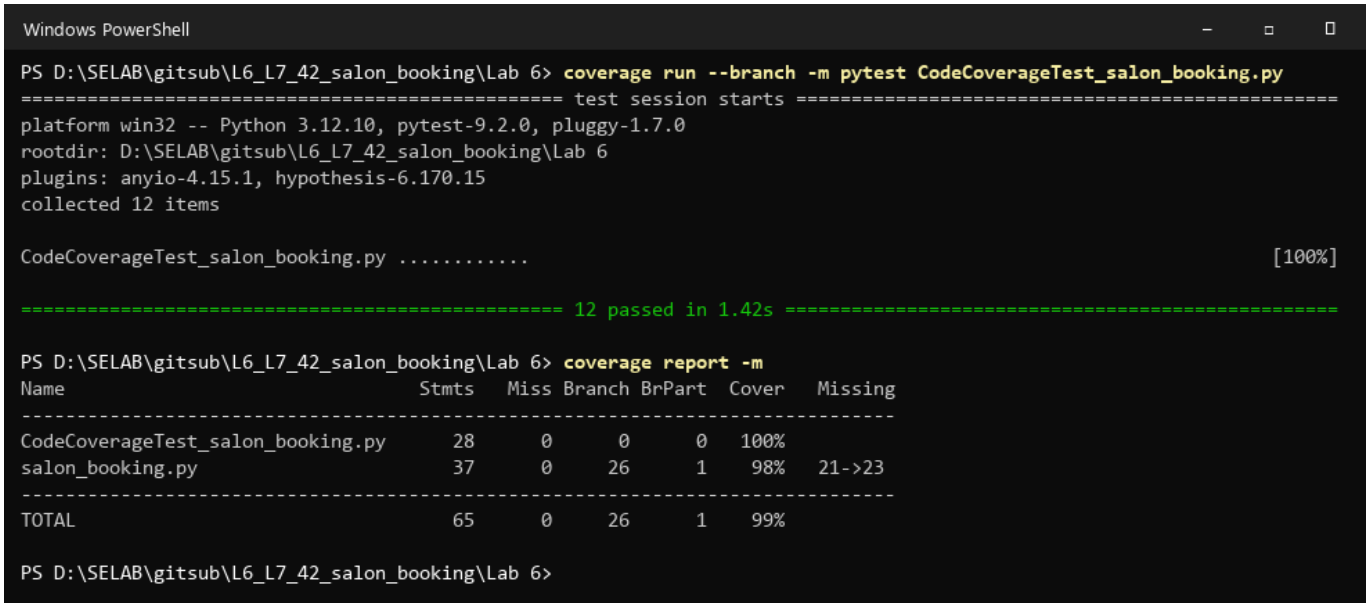
The code coverage tests execute every line, but never take the path where a long overrun (extra > 60 min) is done by a non-senior stylist, so the decision at line 21 is never False. The branch coverage tests drive every if/elif/else to both outcomes and add boundary values (duration 0, extra exactly 0 and 60, walk-in party of 2 and 3, party of 4 with 2 and 3 services, and each arrival cutoff tested at cutoff, cutoff-1, cutoff-15 and cutoff-16 for all colour/weekend combinations).

Commands: `coverage run -m pytest <file>`, `coverage run --branch -m pytest <file>`, `coverage report -m`

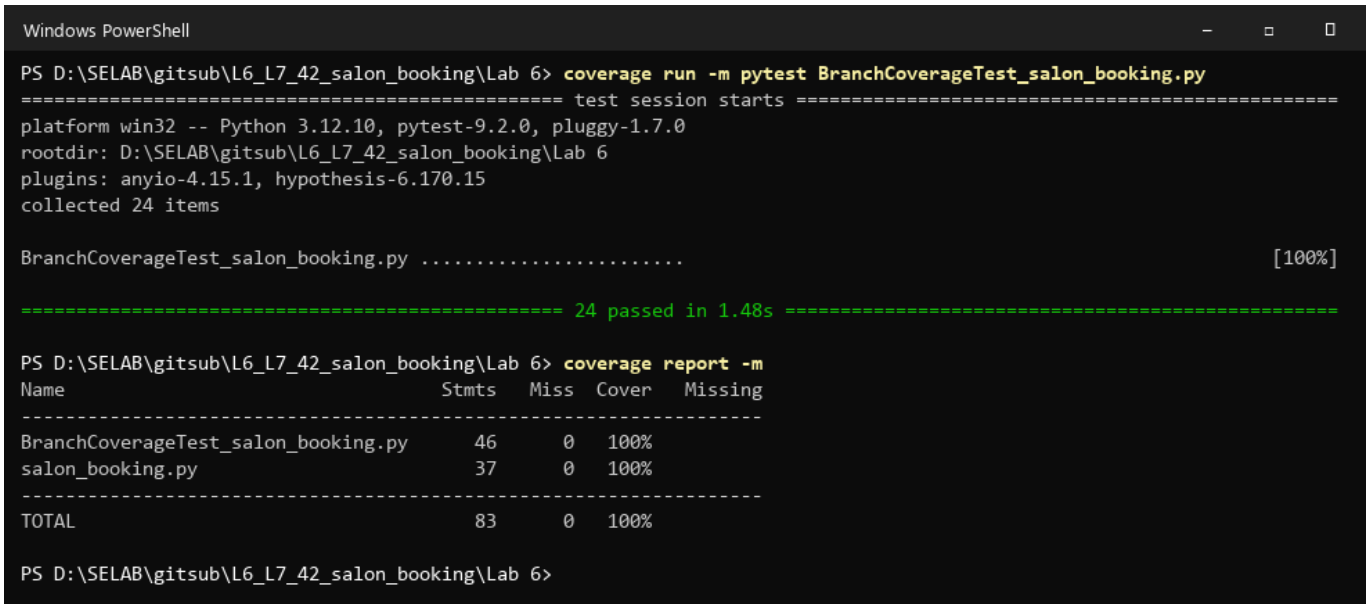
1. Code coverage test cases - line coverage



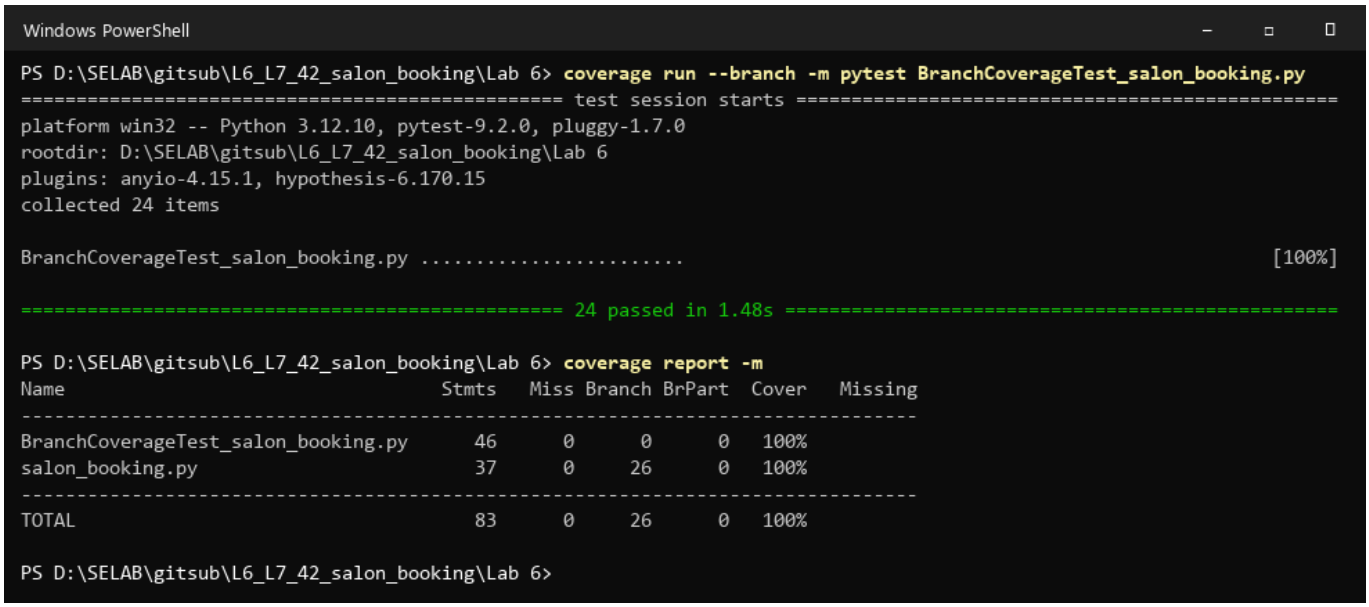
2. Code coverage test cases - branch coverage


Windows PowerShell
PS D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 6> coverage run --branch -m pytest CodeCoverageTest_salon_booking.py
========== test session starts ==========
platform win32 -- Python 3.12.10, pytest-9.2.0, pluggy-1.7.0
rootdir: D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 6
plugins: anyio-4.15.1, hypothesis-6.170.15
collected 12 items
CodeCoverageTest_salon_booking.py ..... [100%]
========== 12 passed in 1.42s ==========
PS D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 6> coverage report -m
Name Stmts Miss Branch BrPart Cover Missing
CodeCoverageTest_salon_booking.py 28 0 0 0 100%
salon_booking.py 37 0 26 1 98% 21->23
TOTAL 65 0 26 1 99%
PS D:\SELAB\gitsub\L6_L7_42_salon_booking\Lab 6>

3. Branch coverage test cases - line coverage



4. Branch coverage test cases - branch coverage



Commits

Commit	Content
Lab 6: add code coverage test cases (100% line coverage)	CodeCoverageTest_salon_booking.py + screenshots
Lab 6: add branch coverage test cases (100% branch coverage)	BranchCoverageTest_salon_booking.py + screenshots